

R Programming and Data Web Extraction: Final Report

Practical examination project · Data extracted 2026-09-18 · R 4.6.1 · Scripts: `main_analysis.R` (analysis) and `build_report.R` (this report)

1. Introduction

This project extracts real-world country data from the **World Bank World Development Indicators (WDI)** through its public web API, then cleans, analyses and visualises it in a single reproducible R script. The data combine each economy's World Bank **region** and **income group** (categorical variables) with 7 numerical indicators for 2022: GDP per capita, life expectancy, population, urbanisation, fertility, health spending and CO₂ emissions.

The source was chosen because it is public, needs no login, is maintained by an official institution, exposes a stable documented API that R can query directly, and supports every part of the examination: over 200 observations, several continuous variables for correlation and histograms, and meaningful grouping variables for bar charts and boxplots.

The research question is: **How does life expectancy vary between countries, and how strongly is it associated with national income?** The analysis covers the overall distribution of life expectancy, differences between regions and income groups (with ANOVA), the correlation and regression between GDP per capita and life expectancy, and outliers and their influence.

Pipeline. `main_analysis.R` performs the full workflow (website → extraction → cleaning → analysis → visualisation → export → automated summary) and writes machine-readable results to `output/`. `build_report.R` runs that script and turns its results into this PDF, so every number here comes from the same run. An optional web dashboard in `frontend/` presents the same exported results interactively.

2. Data Extraction

Table 1. Source and extraction details

Item	Detail
Website	World Bank Open Data - World Development Indicators (API v2)
API base URL	https://api.worldbank.org/v2/
Country endpoint	https://api.worldbank.org/v2/country?format=json&per_page=400
Indicator endpoint	<a href="https://api.worldbank.org/v2/country/all/indicator/<CODE>?format=json&date=2022&per_page=400">https://api.worldbank.org/v2/country/all/indicator/<CODE>?format=json&date=2022&per_page=400
Extraction date	2026-09-18
Reference year	2022
Why this year	2022 = best joint coverage of all seven indicators among the 2021-2023 vintages when the source was selected
Method	HTTP GET of public JSON endpoints (base R <code>url()</code>) parsed with <code>jsonlite::fromJSON()</code>
Error handling	Each request is retried up to 3 times with a growing pause; empty responses and World Bank error messages stop the run with an explanation
R packages	<code>jsonlite</code> , <code>dplyr</code> , <code>tidyr</code> , <code>stringr</code> , <code>readr</code> , <code>ggplot2</code> , <code>scales</code>
Observations collected	295 economies (217 countries + 78 aggregates)
Variables collected	19 columns (10 metadata fields, 7 indicators, year, extraction date)

Extraction used 8 HTTP GET requests: one to the country endpoint (names, ISO codes, region, income group, lending type, capital city and coordinates) and one per indicator for 2022. Each JSON response was checked for World Bank error messages and empty results, then parsed with `jsonlite::fromJSON()`. The indicator records were reshaped from long to wide format with `tidyr::pivot_wider()` and joined to the country metadata by ISO3 code. The raw joined table (295 rows × 19 columns) was saved unchanged as `web_extracted_data.csv`.

Table 2. Indicators extracted

API code	Clean name	Meaning	Unit
NY.GDP.PCAP.CD	<code>gdp_per_capita_usd</code>	GDP per capita	current US\$
SP.DYN.LE00.IN	<code>life_expectancy_years</code>	Life expectancy at birth, total	years
SP.POP.TOTL	<code>population</code>	Total population	people
SP.URB.TOTL.IN.ZS	<code>urban_pop_pct</code>	Urban population as a share of total population	percent (0-100)
SP.DYN.TFRT.IN	<code>fertility_rate</code>	Total fertility rate	births per woman
SH.XPD.CHEX.PC.CD	<code>health_exp_per_capita_usd</code>	Current health expenditure per capita	current US\$
EN.GHG.CO2.PC.CE.AR5	<code>co2_per_capita_t</code>	CO2 emissions per capita, excluding land use (LULUCF)	tonnes CO2e per person

3. Data Cleaning

Inspection of the raw data (`head()`, `str()`, `summary()`, `dim()`, `names()`) found: **78 aggregate rows** (e.g. 'World', 'Euro area', 'High income') mixed in with countries; category labels with **trailing spaces** (90 region values and 4 country names, e.g. 'Sub-Saharan Africa '); **empty strings** instead of missing values in the admin-region, capital-city and coordinate fields; **coordinates stored as text**; and **missing indicator values**.

There were **0 duplicate rows** and **0 duplicate ISO3 codes**. No currency symbols, percentage signs or thousands separators appeared in the numeric fields, but the conversion step strips them defensively.

To make the cleaning stage reproducible from the saved raw file, it re-reads `web_extracted_data.csv` with every column as text and applies the steps below.

Table 3. Cleaning log (rows remaining after each step)

Step	Rows after	Detail
0. Raw data read	295	All columns read as character
1. Trim spaces / empty -> NA	295	Removed trailing spaces (e.g. 'Sub-Saharan Africa '); blank cells set to NA
2. Remove aggregates	217	78 aggregate rows removed (e.g. 'World', 'Euro area')
3. Remove duplicates	217	0 duplicate rows removed
4. Remove unnecessary columns	217	Dropped: iso2, admin_region, lending_type, capital_city, longitude, latitude, data_year, extraction_date
5. Standardise names	217	API codes renamed to descriptive snake_case names
6. Convert data types	217	7 indicators -> numeric; region & income_level -> factor; 'Not classified' income -> NA
7. Handle missing values	209	8 countries lacking GDP per capita or life expectancy removed; other NAs retained (no imputation)
8. Derive variables	209	Added population_millions and log10_gdp_per_capita

Key cleaning decisions

- **Aggregates removed:** regional and income-group totals are not countries; keeping them would double-count and distort every statistic.
- **Removed variables:** iso2, admin_region, lending_type, capital_city, longitude, latitude, data_year, extraction_date. They are identifiers or location metadata that are irrelevant to the analysis, or constant run information.
- **Missing values:** 8 countries with no 2022 GDP-per-capita value (Cuba; Eritrea; Gibraltar; St. Martin (French part); Korea, Dem. People's Rep.; South Sudan; British Virgin Islands; Yemen, Rep.) were removed because GDP per capita and life expectancy are the core analysis variables, and imputing a specific country's income would fabricate data. Missing values in secondary indicators (health spending, CO₂) were **retained** as NA and excluded pairwise with `na.rm = TRUE`, so no rows were lost for them.
- **Type conversion:** the seven indicators were converted from text to numeric with `as.numeric()` after stripping any `,` `$` `%` or spaces; region and income group became factors, with income ordered Low → High.
- **Name standardisation:** API codes such as `NY.GDP.PCAP.CD` were renamed to snake_case names such as `gdp_per_capita_usd`. The full data dictionary is in `output/data_dictionary.csv`.
- **Transformed variables:** `population_millions` and `log10_gdp_per_capita` were added. GDP per capita is heavily right-skewed, so the log scale is used for correlation, regression and plotting.

Table 4. Missing values after aggregates were removed vs. in the final cleaned data

Variable	Missing after aggregates removed	Missing in final data
gdp_per_capita_usd	8	0
health_exp_per_capita_usd	26	21
co2_per_capita_t	14	12

The final cleaned dataset, `cleaned_data.csv`, has **209 countries and 13 variables** (4 identifier/categorical and 9 numeric). 33 of 2717 cells (1.21%) are missing, all in secondary indicators.

3.1 Automated data quality assurance

After cleaning, the script runs **29 automated checks** and stops with an explanation if any fail. They cover dataset size, required columns, ISO3 uniqueness, duplicates, aggregate removal, data types, missingness, plausibility ranges, agreement between the saved CSVs and the analysed data, and the presence of every output file. The plausibility ranges are definitional limits, e.g. urban population must be 0–100%; statistical outliers are handled separately in Section 4.4. Result: **PASS**, with 28 of 28 completed checks passed. The final check, that this PDF exists, is completed by `build_report.R` once the PDF is written. The full list is in `ouput/qa_report.json`.

Table 5. QA checks by category

Category	Checks	Passed	Failed	Pending
Aggregates	1	1	0	0
Export	2	2	0	0
Missingness	2	2	0	0
Outputs	8	7	0	1
Plausibility	7	7	0	0
Size	2	2	0	0
Statistics	1	1	0	0
Structure	1	1	0	0
Types	2	2	0	0
Uniqueness	3	3	0	0

4. Analysis

4.1 Descriptive statistics

Table 6. Descriptive statistics for 6 numerical variables (n = non-missing countries)

Variable	N	Missing	Mean	Median	Min	Max	SD
gdp_per_capita_usd	209	0	21,200.87	7,655.59	301.83	226,052.06	31,131.45
life_expectancy_years	209	0	73.11	74.21	18.82	85.75	7.94
urban_pop_pct	209	0	62.41	63.48	14.69	100.00	23.13
fertility_rate	209	0	2.40	1.95	0.68	6.25	1.22
health_exp_per_capita_usd	188	21	1,388.27	450.24	15.86	12,586.20	2,233.18
co2_per_capita_t	197	12	4.62	2.59	0.00	78.73	7.70

Life expectancy averages **73.1 years** (median 74.2, SD 7.9), ranging from 18.8 years (Central African Republic) to 85.7 years (Monaco). GDP per capita is extremely unequal: the mean (US\$21,201) is 2.8 times the median (US\$7,656), a strong right skew driven by a small number of very rich economies. Fertility averages 2.40 births per woman, and CO₂ per capita is also right-skewed (mean 4.62 t vs. median 2.59 t).

4.2 Group analysis and ANOVA

Table 7. Summary by World Bank region

Region	Countries	Mean life exp.	Median life exp.	Median GDP pc (US\$)	Mean fertility	Mean urban %
North America	3	80.20	81.09	78,008.69	1.46	87.46
Europe & Central Asia	57	78.36	78.77	24,946.57	1.64	67.37
Middle East, North Africa, Afghanistan & Pakistan	22	75.86	76.73	4,857.36	2.46	77.15
Latin America & Caribbean	39	74.48	74.49	11,405.79	1.80	68.69
South Asia	6	74.48	73.51	3,029.56	1.87	39.12
East Asia & Pacific	36	73.59	72.27	6,682.84	2.23	62.24
Sub-Saharan Africa	46	63.10	63.58	1,341.51	4.11	45.44

Table 8. Summary by World Bank income group

Income group	Countries	Mean life exp.	SD life exp.	Median GDP pc (US\$)	Mean fertility
Low income	21	61.06	10.58	827.33	4.69
Lower middle income	47	67.69	5.19	2,325.03	3.23
Upper middle income	58	73.27	3.58	6,653.46	2.07
High income	83	79.11	4.27	35,653.90	1.59

Mean life expectancy ranges from **80.2 years in North America** to **63.1 years in Sub-Saharan Africa**. A one-way ANOVA of life expectancy by region gives **F(6, 202) = 34.0, p = 3.4e-28**, so the regional differences are far larger than would be expected by chance. Region accounts for about **50%** of the variation in life expectancy ($\eta^2 = 0.50$). By income group, mean life expectancy rises at every step, from 61.1 years (low income) to 79.1 years (high income), a gap of **18.0 years**. Fertility moves in the opposite direction (4.69 to 1.59 births per woman). North America contains only 3 countries, so its regional average should be interpreted with care.

Regional SDs range from 2.4 (North America) to 8.1 (Sub-Saharan Africa), so the equal-variance assumption is questionable. Welch's ANOVA, which does not assume equal variances, gives **F(6, 21.8) = 21.8, p = 3.6e-08**, so the conclusion is unchanged.

4.3 Correlation analysis: GDP per capita vs. life expectancy

Table 9. Correlation coefficients (n = 209 countries)

Variable pair	Method	r	Interpretation
GDP per capita vs life expectancy	Pearson	0.57	moderate positive
log10(GDP per capita) vs life expectancy	Pearson	0.79	strong positive
GDP per capita vs life expectancy	Spearman (rank)	0.85	very strong positive

Direction and strength: the relationship is **positive**, so richer countries tend to have longer life expectancy. On the raw dollar scale Pearson $r = 0.570$ (moderate positive) because the relationship is curved: gains in life expectancy flatten as income rises. After a log10 transformation of GDP per capita the

relationship is close to linear and $r = \mathbf{0.792}$ (95% CI 0.735 to 0.837, $p = 3.6e-46$), a strong positive association. The rank-based Spearman coefficient ($\rho = 0.846$, very strong positive) confirms the monotonic relationship without assuming linearity.

Meaning and limits: correlation measures association, not causation. Income is linked to health systems, education, sanitation, conflict and disease burden, any of which could drive both variables. Section 4.6 shows how the coefficient changes when unusual countries are removed.

4.4 Outlier analysis (1.5 × IQR rule)

Table 10. IQR outlier bounds (Q1 - 1.5 IQR, Q3 + 1.5 IQR)

Variable	Q1	Q3	IQR	Lower bound	Upper bound	Outliers	Low	High
gdp_per_capita_usd	2,899.16	28,360.30	25,461.14	-35,292.56	66,552.02	16	0	16
life_expectancy_years	67.71	78.53	10.82	51.49	94.76	1	1	0
fertility_rate	1.51	3.13	1.61	-0.91	5.55	6	0	6

The IQR rule was chosen because it relies on quartiles, which are robust to skewed data, unlike the mean $\pm$ 3 SD rule, which is distorted by the very outliers it is trying to detect.

- **GDP per capita: 16 high outliers** above US\$66,552: Monaco (US\$226,052), Liechtenstein (US\$188,055), Luxembourg (US\$123,720), Bermuda (US\$119,969), Norway (US\$113,122), Ireland (US\$105,191), Switzerland (US\$97,809), Cayman Islands (US\$92,938), Singapore (US\$91,228), Qatar (US\$88,701), Isle of Man (US\$88,329), United States (US\$78,009), Iceland (US\$76,377), Denmark (US\$67,781), Channel Islands (US\$67,627), Faroe Islands (US\$67,069). These are real, very wealthy (mostly small) economies; they reflect the right skew rather than errors.
- **Life expectancy: 1 low outlier** below 51.5 years: **Central African Republic at 18.8 years**. This is the value published by the World Bank, but the same API shows the country's series swinging sharply from year to year (51.9 in 2015, 51.0 in 2016, 45.2 in 2017, 52.3 in 2018, 31.5 in 2019, 50.6 in 2020, 40.3 in 2021, 18.8 in 2022, 57.4 in 2023), which suggests a crisis-mortality estimate rather than an extraction error. It is flagged as a data-quality caveat rather than deleted.
- **Fertility rate: 6 high outliers** above 5.55 births per woman, all in Sub-Saharan Africa: Somalia, Fed. Rep. (6.25), Chad (6.21), Niger (6.13), Congo, Dem. Rep. (6.11), Central African Republic (6.02), Mali (5.69).

No outliers were removed. They are genuine published observations, and removing them would misrepresent global inequality; their influence is measured instead (Sections 4.5 and 4.6).

4.5 Regression analysis (additional)

To quantify the relationship, an ordinary least squares model `life_expectancy_years ~ log10_gdp_per_capita` was fitted to all 209 countries. GDP per capita is strongly right-skewed; log10 makes the relationship approximately linear and the slope reads as years per ten-fold difference in GDP per capita.

Table 11. Regression coefficients (OLS)

Term	Estimate	Std. error	t	p-value	95% CI low	95% CI high
Intercept	34.42	2.10	16.37	3.5e-39	30.27	38.56
log10(GDP per capita)	9.86	0.53	18.64	3.6e-46	8.82	10.91

Result: the slope of **9.86** (SE 0.53, $t = 18.6$, $p = 3.6e-46$; 95% CI 8.82 to 10.91) means that a ten-fold difference in GDP per capita is associated with about 9.9 more years of life expectancy. The model explains

63% of the between-country variation ($R^2 = 0.627$, adjusted $R^2 = 0.625$; residual SE 4.86 years; $F(1, 207) = 347.4$).

Association, not causation. The slope describes how life expectancy differs between richer and poorer countries; it does not show that raising a country's GDP would raise its life expectancy by that amount. Healthcare access, education, public infrastructure, demographics, environmental conditions and institutional factors are plausible confounders that are related to both variables and are not in the model. They are possible explanations, not tested mechanisms.

Diagnostics (Figure 5): Adding a squared log-GDP term improves the fit (F-test $p = 0.024$), so some curvature remains even on the log scale; the straight line is a useful summary rather than an exact form. Residuals depart from normality (Shapiro-Wilk $p = 1.1e-15$; skewness -3.31, a longer tail of countries far BELOW the fitted line), so p-values and confidence intervals should be read as approximate. With $n = 209$ the coefficient estimates themselves remain informative. 8 countries exceed the Cook's distance screening value of $4/n = 0.0191$, 13 have high leverage ($> 2p/n$) and 1 has a standardised residual beyond ± 3 .

Table 12. Potentially influential countries (Cook's distance $> 4/n$)

Country	Life exp.	Std. residual	Leverage	Cook's D
Central African Republic	18.818	-8.723	0.023	0.910
Nigeria	54.079	-2.990	0.007	0.033
Chad	54.528	-1.934	0.015	0.029
Greenland	71.230	-2.068	0.013	0.028
Somalia, Fed. Rep.	53.931	-1.598	0.021	0.027
Syrian Arab Republic	72.790	1.769	0.014	0.023
Naoero	61.930	-2.838	0.006	0.023
Afghanistan	65.617	1.254	0.027	0.022

4.6 Sensitivity / robustness analysis

The correlation and regression were re-estimated after removing each group of unusual countries in turn, to test whether the conclusion rests on a few observations.

Table 13. Relationship between log GDP per capita and life expectancy under different exclusions

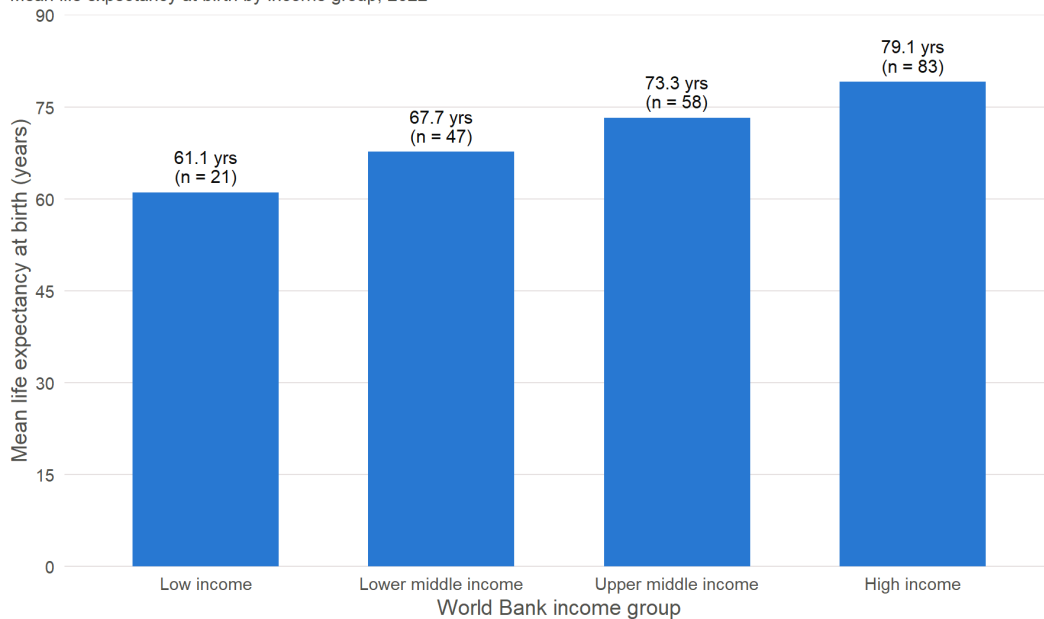
Scenario	n	Excluded	Pearson r (log GDP)	Spearman rho	Slope (yrs per 10x GDP)	R-squared
All countries (baseline)	209	none	0.792	0.846	9.864	0.627
Excluding life-expectancy IQR outliers	208	Central African Republic	0.834	0.844	9.227	0.695
Excluding GDP-per-capita IQR outliers	193	16 high-income economies	0.766	0.816	10.334	0.588
Excluding influential points (Cook's D $> 4/n$)	201	8 countries	0.859	0.859	9.198	0.737

Across the 4 scenarios Pearson r (log GDP) ranges from 0.766 to 0.859 (largest change from the baseline 0.792: 0.067) and the slope from 9.2 to 10.3 years per ten-fold GDP. The direction is unchanged but the strength label varies (strong positive / very strong positive), so the size of the association depends partly on unusual countries.

5. Visualisation

Average life expectancy rises with each World Bank income group

Mean life expectancy at birth by income group, 2022



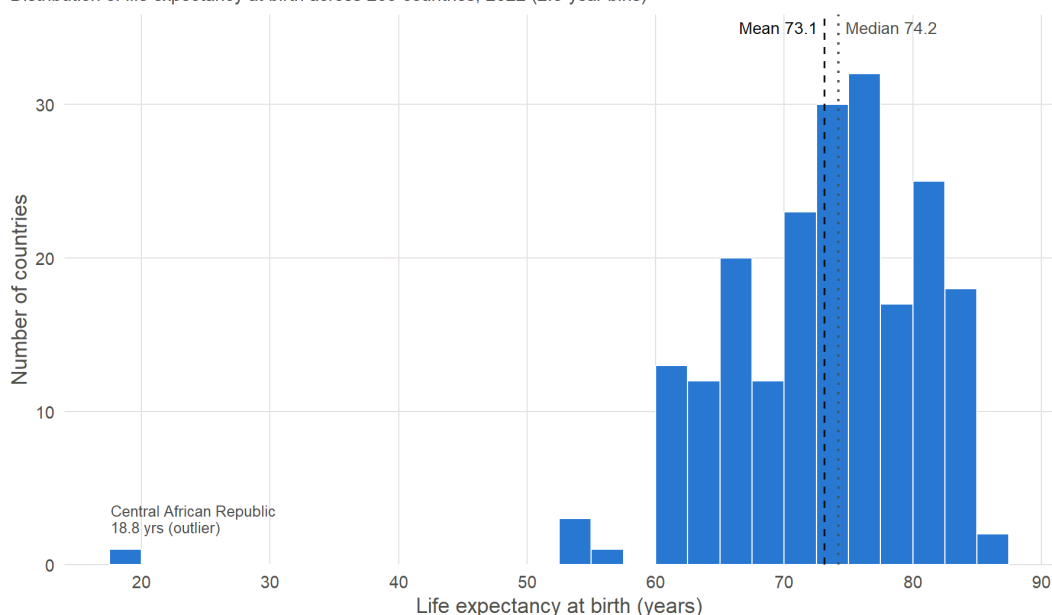
Source: World Bank World Development Indicators API, 2022 data (extracted 2026-09-18)

Figure 1. Bar chart: Mean life expectancy by World Bank income group

Mean life expectancy rises at every step with income group, from 61.1 years in low-income countries to 79.1 years in high-income countries, a gap of 18 years. Each step up the income ladder adds between 5.6 and 6.6 years on average. The low-income group is the smallest (n = 21) and most variable (SD 10.6 years), so its mean is the least precise.

Left-skewed: 49% of countries have a life expectancy of 70-80 years

Distribution of life expectancy at birth across 209 countries, 2022 (2.5-year bins)



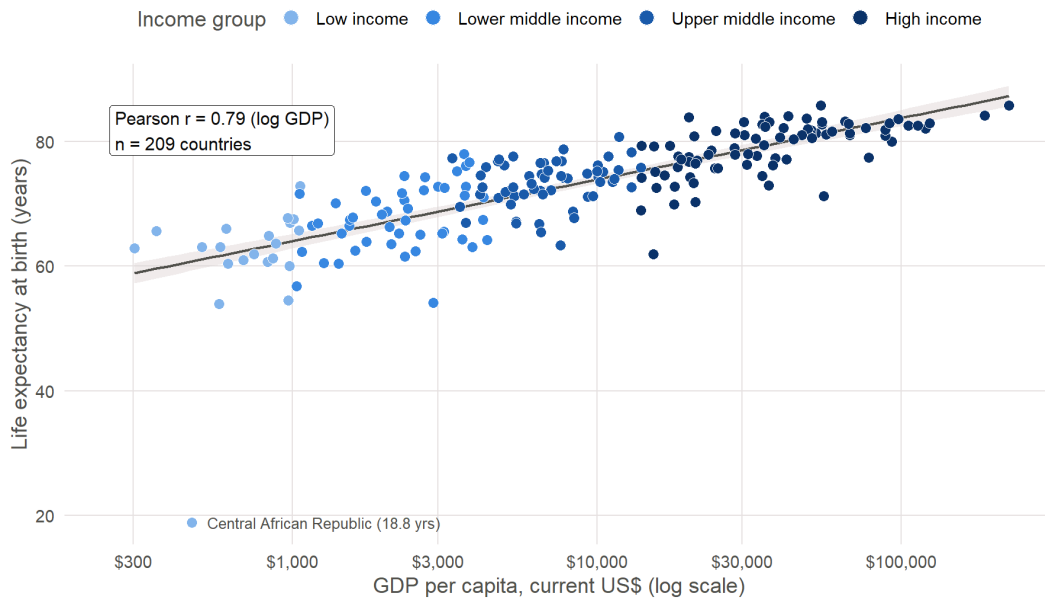
Source: World Bank World Development Indicators API, 2022 data (extracted 2026-09-18)

Figure 2. Histogram: Distribution of life expectancy at birth

The distribution is left-skewed: 49% of countries fall between 70 and 80 years, and the mean (73.1) sits below the median (74.2). Only 5 countries are below 60 years; the isolated bar at 18.8 years is Central African Republic, the outlier discussed in the outlier analysis.

Richer countries tend to have longer life expectancy

GDP per capita (log scale) vs life expectancy at birth, 2022; line = linear fit on log GDP



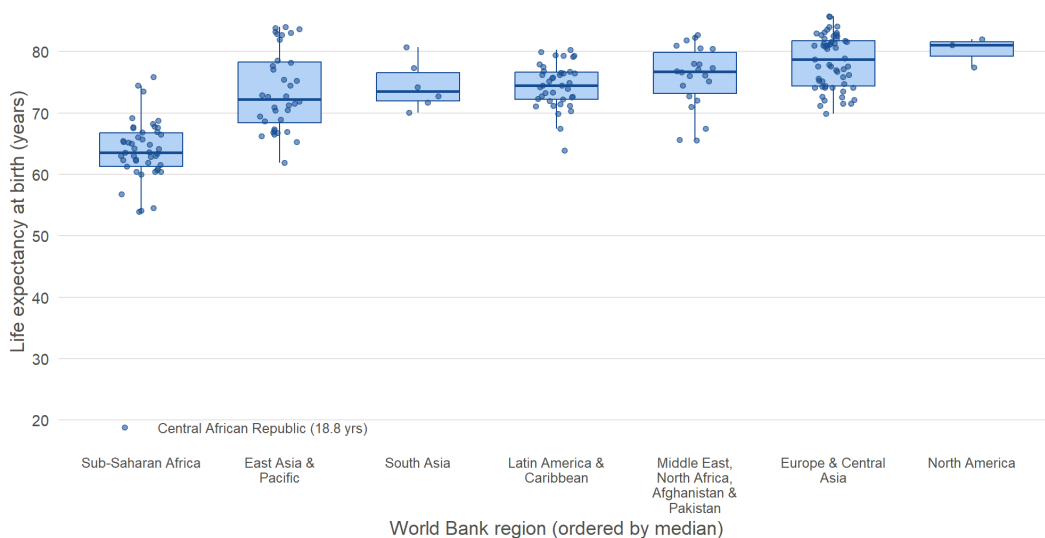
Source: World Bank World Development Indicators API, 2022 data (extracted 2026-09-18)

Figure 3. Scatter plot: GDP per capita (log scale) vs life expectancy

Points rise from lower left to upper right, a strong positive association ($r = 0.79$ on log GDP), and the income-group colours follow the same gradient. Scatter around the line is widest in the low income group (residual SD 10.2 years), so income alone predicts life expectancy least well there. Some countries sit well below the line for their income, e.g. Naoero, a high-income economy at 61.9 years.

Sub-Saharan Africa has the lowest life expectancy of any region

Life expectancy at birth by World Bank region, 2022 (points = countries; boxes = median and IQR)



Source: World Bank World Development Indicators API, 2022 data (extracted 2026-09-18)

Figure 4. Boxplot: Life expectancy by World Bank region

Sub-Saharan Africa's median life expectancy (63.6 years) is about 9 years below the next-lowest region (East Asia & Pacific, 72.3 years), and its whole box sits below every other region's box. East Asia & Pacific shows the widest interquartile range (9.9 years). North America has only 3 countries, so its box should be read with care.

Figures 1–4 are the four required visualisations. Figure 5 is an additional diagnostic figure supporting the regression in Section 4.5.

Figure 5 (additional). Diagnostics for life_expectancy ~ log10(GDP per capita), n = 209

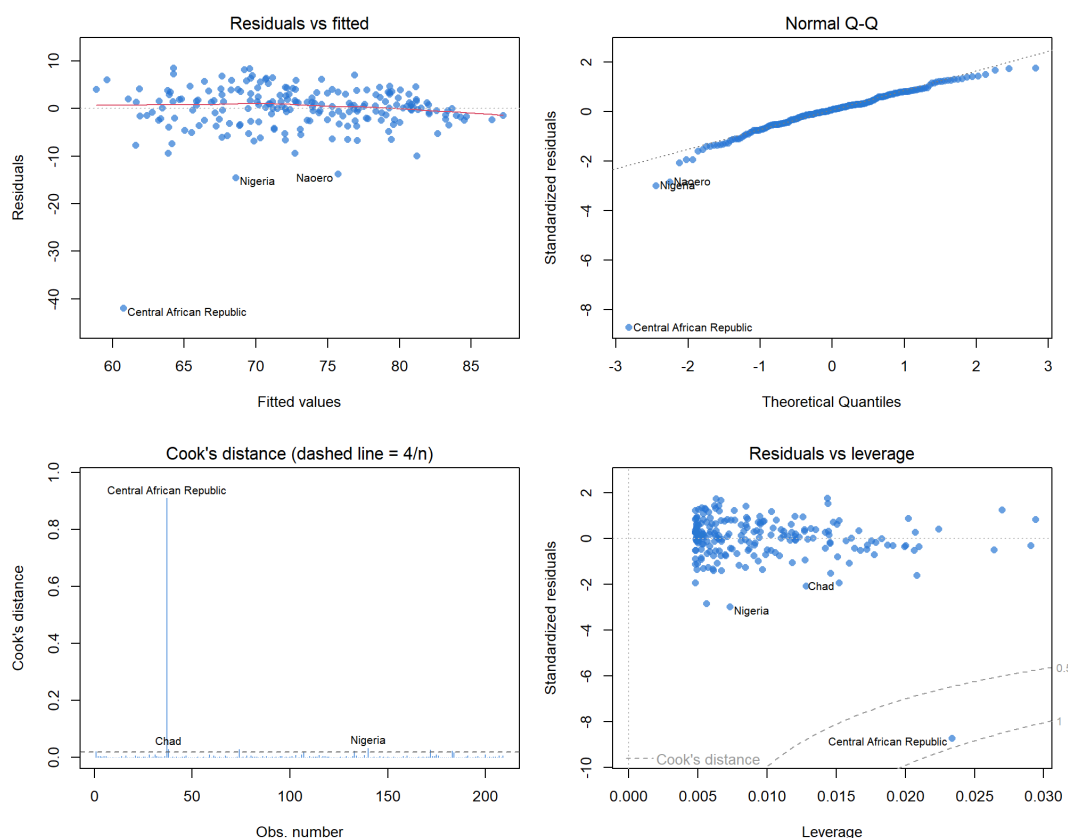


Figure 5. Regression diagnostics: Model assumptions and influential countries (additional)

Adding a squared log-GDP term improves the fit (F-test $p = 0.024$), so some curvature remains even on the log scale; the straight line is a useful summary rather than an exact form. Residuals depart from normality (Shapiro-Wilk $p = 1.1e-15$; skewness -3.31, a longer tail of countries far BELOW the fitted line), so p-values and confidence intervals should be read as approximate. With $n = 209$ the coefficient estimates themselves remain informative. 8 countries exceed the Cook's distance screening line ($4/n$), led by Central African Republic, Nigeria, Chad; removing them changes r from 0.792 to 0.859 (see the sensitivity analysis).

6. Limitations

- **Cross-sectional design.** The analysis uses one year (2022) of country-level data, so it shows how countries differ at a point in time; it cannot establish temporal ordering or causal effects.
- **Country-level (ecological) inference.** Relationships between country averages do not necessarily hold for individuals within countries, and small territories count equally with populous countries.
- **Missing data.** 8 countries were removed because GDP per capita or life expectancy was missing (Cuba; Eritrea; Gibraltar; St. Martin (French part); Korea, Dem. People's Rep.; South Sudan; British Virgin Islands; Yemen, Rep.). Their income groups are: Low income 4, Upper middle income 1, High income 3, so the missing cases are not a random sample and the cleaned data may under-represent some settings (50% of the dropped countries are low income vs 10% of the analysed countries). 21 health-expenditure and 12 CO2 values are also missing and were excluded pairwise.
- **Confounding.** GDP per capita and life expectancy are both linked to healthcare access, education, public infrastructure, demographics, environmental conditions and institutional quality. These are possible explanations for the association, not mechanisms tested here.

- **Outliers and influential observations.** Extreme observations - notably Central African Republic at 18.8 years - can shift correlations and regression estimates. The sensitivity analysis shows how much: Across the 4 scenarios Pearson r (log GDP) ranges from 0.766 to 0.859 (largest change from the baseline 0.792: 0.067) and the slope from 9.2 to 10.3 years per ten-fold GDP. The direction is unchanged but the strength label varies (strong positive / very strong positive), so the size of the association depends partly on unusual countries.
- **Source revisions.** World Bank WDI values are revised as new estimates arrive. Re-running the extraction later may give slightly different numbers than the 2026-09-18 extraction reported here; the code, source, year and extraction date are recorded so the method is reproducible.

7. Conclusion

The pipeline accessed a public web API, extracted and saved the raw data, cleaned it with documented decisions, checked it with automated QA, and ran descriptive, group (ANOVA), correlation, regression, outlier and sensitivity analyses. It produced four required figures and one diagnostic figure, and exported the cleaned dataset with machine-readable results. The automated summary, generated from the calculated results, reads:

- Data: 295 economies were extracted from the World Bank API on 2026-09-18; after removing 78 aggregates and 8 countries missing core values, 209 countries and 13 variables remain.
- Missingness: 33 of 2717 cells (1.21%) are missing in the cleaned data, all in `health_exp_per_capita_usd` (21) and `co2_per_capita_t` (12); core variables are complete.
- Life expectancy (2022): mean 73.1 years, median 74.2, ranging from 18.8 (Central African Republic) to 85.7 (Monaco), SD 7.9.
- GDP per capita: mean US\$21,201 but median only US\$7,656 - a strongly right-skewed distribution (min US\$302 in Burundi, max US\$226,052 in Monaco).
- Groups: mean life expectancy is highest in North America (80.2 yrs) and lowest in Sub-Saharan Africa (63.1 yrs); high-income countries average 79.1 yrs vs 61.1 yrs for low-income countries (gap 18.0 yrs). ANOVA across regions: $F(6, 202) = 34.0$, $p = 3.4e-28$, eta-squared = 0.50.
- Correlation: log GDP per capita and life expectancy show a strong positive correlation (Pearson $r = 0.79$, 95% CI 0.73 to 0.84, $p = 3.6e-46$; Spearman $\rho = 0.85$). This is an association, not proof of causation.
- Regression: each ten-fold increase in GDP per capita is associated with 9.9 more years of life expectancy (95% CI 8.8 to 10.9, $p = 3.6e-46$); $R^2 = 0.63$, adjusted $R^2 = 0.62$.
- Outliers (1.5 x IQR rule): 16 high GDP-per-capita outlier(s) above US\$66,552, 1 low life-expectancy outlier(s) below 51.5 years (Central African Republic) and 6 high fertility outlier(s); all were retained.
- Sensitivity: Across the 4 scenarios Pearson r (log GDP) ranges from 0.766 to 0.859 (largest change from the baseline 0.792: 0.067) and the slope from 9.2 to 10.3 years per ten-fold GDP. The direction is unchanged but the strength label varies (strong positive / very strong positive), so the size of the association depends partly on unusual countries.
- Outputs: 4 required figures and 1 additional figure generated; automated data QA PASS (28 of 28 completed checks passed; the final-report check runs once the PDF is built).

Overall: life expectancy shows a strong positive association with national income ($r = 0.79$ on log GDP; about 9.9 years per ten-fold GDP difference), differs markedly by region (lowest in Sub-Saharan Africa) and rises across every income group. The data are skewed (GDP per capita to the right, life expectancy to the left) and contain genuine outliers, which were reported and tested rather than removed. These findings

describe associations between countries in 2022. They do not establish that higher income causes longer lives.

Appendix: Section A results (student dataset)

Table A1. Student dataframe with Grade (10 observations, 6 original variables + Grade)

Student	Gender	Age	Programme	Score	Attendance	Grade
Alice	Female	21	Computer Science	78	92	B
Brian	Male	23	Data Science	65	81	C
Charles	Male	22	Computer Science	55	76	D
Diana	Female	20	Information Systems	88	95	A
Edward	Male	24	Data Science	72	84	B
Faith	Female	21	Computer Science	91	97	A
George	Male	22	Information Systems	48	68	F
Hannah	Female	23	Data Science	83	89	A
Isaac	Male	20	Computer Science	61	79	C
Jane	Female	22	Information Systems	74	86	B

Missing values: 0. Grade rule: 80-100 A, 70-79 B, 60-69 C, 50-59 D, below 50 F.

Table A2. Score statistics

Statistic	Value
Mean	71.50
Median	73.00
Highest	91.00
Lowest	48.00
Standard deviation	14.18

Table A3. Average score by programme

Programme	Students	Average_Score
Data Science	3	73.33
Computer Science	4	71.25
Information Systems	3	70.00

Table A4. Students scoring above the overall average of 71.5

Student	Programme	Score	Grade
Faith	Computer Science	91	A
Diana	Information Systems	88	A
Hannah	Data Science	83	A
Alice	Computer Science	78	B
Jane	Information Systems	74	B
Edward	Data Science	72	B